

SURENDRA BATIKIRI

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PROFESSIONAL SUMMARY

Motivated Computer Science graduate and aspiring Full Stack Developer with hands-on Python programming, SQL database design, and backend development fundamentals. Demonstrated ability to build end-to-end systems through academic projects and a self-built, deployed e-commerce web application spanning machine learning, AI integration, and data-driven application design. Certified in Java Full Stack Development with exposure to web technologies including HTML and CSS. Quick learner with a strong problem-solving mindset, eager to contribute to AI-powered product engineering in a fast-moving startup environment.

CORE COMPETENCIES

Python Development | FastAPI | REST API Development | Backend Engineering | SQL & Database Design | PostgreSQL / MySQL / Oracle | Machine Learning | AI Integration | Data Analysis | Problem Solving | Git & GitHub | Agile Collaboration | Web Development Basics

TECHNICAL SKILLS

Languages: Python, Core Java, C, SQL

Backend: FastAPI, REST API Development, Spring Boot Fundamentals

Frontend: HTML5, CSS3 (basic web development)

Databases: PostgreSQL, MySQL, Oracle

AI / ML: Machine Learning, Deep Learning, Feature Fusion, Emotion Detection

Tools: Git, GitHub, Vercel, MS Excel, MS Word, PowerPoint, Postman

Cloud / DevOps: Linux (OS), Vercel Deployment, Docker (exposure), CI/CD (learning)

Practices: OOP, MVC Architecture, Agile Basics, Problem Solving

PROJECTS

Sri Someshwara Silk Sarees – E-Commerce Web Application

Python · FastAPI · PostgreSQL | Live: sssilksarees.vercel.app

- Independently designed, built, and deployed a full-stack e-commerce web application for a silk sarees retail business, taking it from concept to a live, publicly accessible product.
- Built a RESTful backend using Python and FastAPI to serve product catalog data, handling routing, request validation, and business logic for browsing sarees by category and collection.
- Designed and managed a PostgreSQL database schema to model products, categories, and inventory details, supporting efficient querying for a smooth product browsing experience.
- Deployed and hosted the application end-to-end on Vercel, managing the full solo development lifecycle from database design through backend API development to production release.

Adaptive Feature Fusion Network for Passenger Flow Prediction

Python · Machine Learning · Deep Learning

- Designed and implemented an end-to-end deep learning pipeline to predict metro passenger flow using an Adaptive Feature Fusion Network (AFFN), targeting accurate origin-destination (OD) analysis for large-scale transit systems.
- Engineered a multi-source feature fusion architecture that integrates spatial and temporal transit data streams, improving prediction accuracy over baseline models through advanced fusion strategies.
- Developed data preprocessing and transformation modules in Python to clean, normalize, and structure large-scale passenger flow datasets for model training and evaluation.
- Optimized model performance through hyperparameter tuning and evaluation across standard regression metrics, demonstrating the system's potential for real-world deployment in smart city transit planning.

Emotion-Based Music Recommendation System

Python · Machine Learning · NLP · API Integration

- Built an intelligent music recommendation system that detects user emotions from input signals and maps them to curated playlists, delivering a personalized and context-aware listening experience.
- Implemented an emotion detection module using machine learning classification techniques, processing user input to identify emotional states with actionable accuracy across multiple emotion categories.
- Designed a recommendation engine that dynamically maps detected emotion labels to music genre and playlist categories, enabling real-time, mood-driven song suggestions through API-based music service integration.
- Developed the full application pipeline in Python — from input ingestion and emotion classification to playlist generation and output delivery — following a clean modular architecture.

EDUCATION

B.Tech – Computer Science & Engineering 2022 – 2025

Siddhartha Institute of Engineering and Technology, Hyderabad | 65.45%

Diploma – Mechanical Engineering 2022

Government Polytechnic College, Vaddepally | 60%

SSC (10th Grade) 2019

Zilla Parishad High School, Rajoli | 73%

CERTIFICATIONS

- Java Full Stack Development – 6-Month Certification | Pentagon Coaching Center, Bangalore

Covered: Core Java, SQL, HTML, CSS, REST API Fundamentals, and Web Development Concepts

ACHIEVEMENTS & STRENGTHS

- Independently built and deployed a live full-stack e-commerce web application using Python, FastAPI, and PostgreSQL, managing the complete development lifecycle from database design to production deployment.
- Developed 2 additional end-to-end academic projects spanning machine learning and AI-driven recommendation systems, applying production-oriented thinking to research-grade problems.
- Completed a rigorous 6-month Java Full Stack bootcamp, demonstrating commitment to industry-relevant skill development beyond the academic curriculum.
- Multilingual communicator — fluent in English, Telugu, Kannada, and Hindi — enabling effective collaboration across diverse, cross-functional teams.
- Demonstrated quick-learning ability by independently studying Python and ML frameworks alongside core engineering coursework.